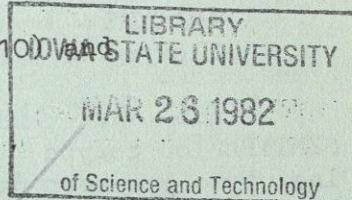


Compiled by the Custodian of the Rothschild Collection of
Siphonaptera, Department of Entomology, British Museum
(Natural History), South Kensington, London SW7 5BD

Please find herewith:

- lists of additional literature for the years 1973 (list 1000) and 1979 (list 3)
- some notes on pulgas vestidas

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Additions to:

NUMBERS OF SPECIES/SUBSPECIES OF SIPHONAPTERA DESCRIBED PER ANNUM [Flea News 5]

YEAR	/	VALID	/	(SYNONYMS)	TOTAL	/	(+SYNONYMS)
correction: 1965		+33		(+3)	= 1796		(2196)
i.e. 1974		+29		(+1)	= 2050		(2463)
1975		+53		(+2)	= 2103		(2518)
1976		+28			= 2131		(2546)
1977		+47		(+4)	= 2178		(2597)
1978		+29		(+1)	= 2207		(2627)
[1979		+30			= 2237		(2657)]

* * *

The word FLEA in some other languages:

AFRIKAANS - vlooi
AMHARIC - kunichā
ARABIC - bargāt
AZERBAJDZHANIAN - bire
BULGARIAN - b'lkha
BYELORUSSIAN - blykha
CATALONIAN - puça
CHINESE - zao, t'iao
CROATIAN - buha
CZECH - blecha
DANISH - loppe
DUTCH - vlo
ENGLISH - flea
FINNISH - kirppu
FRENCH - puce
FRIULANO - pulz, pulç
GERMAN - Floh
GREEK - psullos
HAUSA - k'uma, tunkwiyau
HEBREW - parosh
HUNGARIAN - bolha
ITALIAN - pulce
JAPANESE - nomi
KIKUYU - thuya, thuwa, ngage,
ndutu, kiroboto

KIRGIZIAN - burge
LATIN - pulex
LATVIAN - blusa
LITHUANIAN - blusa
MALAYAN - kutu andjing
MALAYAN (Java) - kēpinjal
MAORI - keha, he pakea nohinohi,
tuiaua, purui, mororohu,
kutu-porenga
MOHAWK - atá:we (dog flea),
otsi:non (human flea)
NORWEGIAN - loppe
PILIPINO - pulgas
POLISH - pchła
PORTUGUESE - pulga
ROMANIAN - purice
RUSSIAN - blokha
SPANISH - pulga
SWEDISH - loppa
TURKISH - pire
UKRAINIAN - blokha
ZULU - i(li)zeze
i(li)khululu, intwakumba (dog
flea)
i(li)tekenya (jigger)

* * *

MAILING LIST

Addition:

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T Y P E S - The following categories can be encountered when consulting the combined Rothschild/Brit.Mus.Nat.Hist. collection of fleas:

HOLOTYPE - the single specimen designated or indicated in the original description as being the type-specimen of a nominal species-group taxon; if this taxon is based on a single specimen, that specimen is the holotype.

ALLOTYPE - a single specimen of the sex opposite to that of the holotype and designated as such in the original description.

PARATYPE - every specimen of a type-series other than the holotype and allotype or neallotype.

SYNTYPE [or cotype] - every specimen of a type-series in which no holotype or lectotype has been designated.

LECTOTYPE - a syntype designated, after the original publication of a species-group name, as being the type-specimen of the taxon bearing that name.

LECTALLOTYPE - a single specimen, preferably a syntype, of the sex opposite to that of the lectotype and designated as such at the time of the lectotype selection.

PARALECTOTYPE - any one of the original syntypes remaining after the selection of a lectotype and lectallotype.

NEOTYPE - a single specimen, preferably from the original type-locality or near, designated as the type-specimen of a nominal species-group taxon of which the holotype or lectotype, and all paratypes, or all syntypes, are lost or destroyed.

TOPOTYPE - a specimen of a nominal species-group taxon from the same locality whence the holotype, lectotype or neotype originated.

These are main categories; two rather subordinate concepts in the 'allotypoid' category are:

NEALLOTYPE - a single specimen of the sex opposite to that of the holotype or neotype, not necessarily belonging to the original type-series, designated later than the original description of the nominal species-group taxon.

NEOLECTALLOTYPE - a single syntype of the sex opposite to that of the lectotype and designated later than the publication of the lectotype selection.

* * *

There are very few syntypes and neotypes in the collection. The various subsequently-selected allotype forms tend to receive little enthusiasm.

* * *

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1973 (list 10)

- Bibikova, V.A. & Gerasimova, N.G. - Experimental feeding of the flea *Xenopsylla nuttalli* Ioff, 1930. Probl. osobo opasn. Infektsii (Saratov) 1(29): 122-125.
- Federman, H.B., Holanda, J.C. de, & Evangelista, A. - Ocorrência de parasitos em gatos (*Felis catus domesticus*) e pombos (*Columba livia*) procedentes de algumas localidades de Minas Gerais. Rev. Patol. trop., Goiânia 2(2): 207-215.
- Gerasimova, N.G. - Some particulars of the reproduction of *Xenopsylla skrjabini* and *X. nuttalli*. Probl. osobo opasn. Infektsii (Saratov) 1(29): 117-121.
- Khamaganov, S.A. - Materials on rodents and their ectoparasites in northern rayons of the Khabarovsk Krai. Vopr. Geogr. Dal'n. Vost. (Khabarovsk) 11: 167-172.
- Kolodenko, A.I. & Zagniborodova, Ye.N. - Fleas of insectivores in Turkmenia and their possible role in the epizootology of plague. Parazity Zhiv. Turkm. (Ashkhabad): 104-111.
- Kozlov, M.P. & Savel'yev, V.N. - On the pathogenicity of bacilli of the thuringiensis group for adults and larvae of fleas. Mater. Konfer. Profilakt. Chumy (Saratov): 215-217.
- Lapina, N.F., Marin, S.N. & Kamnev, P.I. - Notes on fleas of the yellow suslik on the Gornom Mangyshlak. Probl. osobo opasn. Infektsii (Saratov) 6(34): 121-124.
- Lazareva, L.A. & Lazarev, B.V. - On the behaviour of ectoparasites after the death of their hosts in the Gorno-Altai focus of plague. Probl. osobo opasn. Infektsii (Saratov) 2(30): 183-184.
- Mironov, N.P., Nyel'zina, Ye.N., Sorokina, L.Ya. & Bakhtinova, N.Z. - The pathogenicity of spores of the fungus *Beauveria bassiana* (Bals.) Vuill. for fleas in different stages of their ontogeny. Probl. osobo opasn. Infektsii (Saratov) 5(33): 71-75.
- Severova, E.A. & Soldatkin, I.S. - Seasonal dynamics in numerosity of fleas of the genus *Xenopsylla* in Kyzylkumy and Karakumy. Probl. osobo opasn. Infektsii (Saratov) 1(29): 110-116.
- Shvarts, Ye.A. & Bibikov, D.I. - Fleas of marmots as vectors and reservoirs of the plague microbe. Prirodn. Ochagi Chumy Surkov S.S.S.R.: 33-48.
- Zagniborodova, Ye.N., Zabegalova, M.N., Kochkareva, A.V., Avakov, S.M., Sageyev, M.T., Zuichenko, A.N. & Guvva, L.A. - Fleas of the yellow suslik on the Krasnovodsk plateau. Probl. osobo opasn. Infektsii (Saratov) 1(29): 18-25.
- Zagniborodova, Ye.N., Zaitseva, V.I., Kitsayeva, A.Ye., Kochkareva, A.V. & Obrikas, R.G. - The ecologic-faunistic and epizootological characteristics of fleas of south-eastern Kara-Kumy. Probl. osobo opasn. Infektsii (Saratov) 6(34): 47-57.

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- Anonymous - Årsberetning 1978 Statens Skadedyrlaboratorium. Lyngby. (Siph.: 19).
- Anonymous - Fleabite allergic dermatitis in dogs. Canine Pract. 6(2): 63-66.
- Bai, M.K. & Prasad, R.S. - Role of dietary components in vitellogenesis in the rat fleas *Xenopsylla cheopis* and *X. astia* (Siphonaptera). Ent. exp. & appl. 26(1): 80-84.
- Bai, M.K. & Prasad, R.S. - Influence of nutrition on maturation of male rat fleas *Xenopsylla cheopis* and *X. astia* (Siphonaptera: Pulicidae). J. med. Ent. 16(2): 164-165.
- Bosman, B.T. - Hinderlijke of schadelijke mijten en insekten in en om gebouwen in 1978. Rat. & Muis 27(3): 39-41.
- Britt, D.P. & Molyneux, D.H. - Parasites of grey squirrels (*Sciurus carolinensis* Gmelin) in Delamere Forest, Cheshire. Trans. R. Soc. trop. Med. Hyg. 73(3): 326.
- Britt, D.P. & Molyneux, D.H. - Parasites of grey squirrels in Cheshire, England. J. Parasit. 65(3): 408.

- Campos, E.G. & Stark, H.E. - A revaluation of the *Hystrichopsylla occidentalis* group, with description of a new subspecies (Siphonaptera: Hystrichopsyllidae). J. med. Ent. 15(5-6): 431-444, figs. 1-60.
- Chamberlain, W.F. - Methoprene and the flea. Pest Control 47(6): 22-24, figs. 1-26.
- Coulanges, P. - Service de la peste. Arch. Inst. Pasteur Madagascar 46(2): 671-690.
- Karpov, A.A. & Korneyev, G.A. - Behavioural patterns of the great gerbil *Rhombomys opimus* (Rodentia, Cricetidae) during plague epizootics under conditions of variation in population density. Zool. Zh. 58(6): 890-895.
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- Lavoipierre, M.M.J., Radovsky, F.J. & Budwiser, P.D. - The host-parasite interaction of *Peromyscus maniculatus* and the embedded female of *Tunga monositus* (Siphonaptera: Tungidae). J. med. Ent. 16(2): 85-94, figs. 1-15.
- Méndez, E. & Lemke, T.O. - Description of a new species of bat flea from Colombia (Siphonaptera: Ischnopsyllidae). Proc. ent. Soc. Wash. 81(4): 657-662, figs. 1-7.
- Mitchell, R.M. - A list of ectoparasites from Nepalese mammals, collected during the Nepal ectoparasite program. J. med. Ent. 16(3): 226-233.
- Myel'nikova, V.I. - New data on the distribution of the flea *Ornithophaga anomala* Mikulin, 1956. In: Chlenistonogiya i Gelminty (Novosibirsk): 118.
- Nikitina, N.A. & Nikolayeva, G.A. - A study of the ability of some rodents to free themselves of fleas. Zool. Zh. 58(6): 931-933.
- Nikolayev, N.I. - On the history of plague prophylaxis in the U.S.S.R. Zh. Mikro-biol., Moskva (4): 110-115.
- Piotrowski, F. - Ectoparasites of dogs and cats, their importance and distribution in Europe. Wiad. Parazyt. 25(4): 387-397, figs. 1-2.
- Rödl, P. - Investigation on the transfer of fleas among small mammals using radioactive phosphorus. Folia parasit. 26(3): 265-274, figs. 1-7.
- Rutledge, L.C., Moussa, M.A., Zeller, B.L. & Lawson, M.A. - Field studies of reservoirs and vectors of sylvatic plague at Fort Hunter Liggett, California. J. med. Ent. 15(5-6): 452-458, figs. 1-4.
- Sleeman, D.P. - Records of fleas (Siphonaptera) from some Ulster mammals. Ent. Gaz. 30(3): 215-218.
- Vashchyonok, V.S. & Avtushenko, L.A. - Changes in abundance of the causative agent of intestinal yersiniosis (*Yersinia enterocolitica*) in the flea *Xenopsylla cheopis* (Aphaniptera) during the process of blood digestion. Parazitologiya 13(5): 503-509.
- Whitaker, J.O. & Goff, R. - Ectoparasites of wild carnivora of Indiana. J. med. Ent. 15(5-6): 425-430.
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- Yakunin, B.M., Chernova, N.A. & Kunitskaya, N.T. - On the number of generations in the flea *Xenopsylla skrjabini* in Mangyshlak (Aphaniptera). Parazitologiya 13(5): 510-515, figs. 1-2.
- Yates, T.L., Pence, D.B. & Launchbaugh, G.K. - Ectoparasites from seven species of North American moles (Insectivora: Talpidae). J. med. Ent. 16(2): 166-168.
- Zolotova, S.I., Bibikova, V.A. & Murzakhmetova, K. - On the fertility of the flea *Xenopsylla gerbilli minax* (Aphaniptera), parasites of the great gerbil. Parazitologiya 13(5): 497-502.

Best thanks to all who so kindly sent cards
of Greetings and Seasonal Good Wishes

A recently issued illustration of two colourful fleas to accompany
my best wishes to you for 1980 !

* * *

Exactly seventy years ago The Hon. N. Charles Rothschild was offered a fine collection of fleas from Mexico - in those days quite a terra incognita as regards these insects. To his intense disappointment (but surely to NCR's amusement) the collection proved to consist of twelve dressed fleas, known as pulgas vestidas. Two outstanding specimens, each (like the others) about 3 mm tall, are shown on this excellent picture postcard; they actually form one of the most popular exhibits in the public galleries of Tring Museum. Noteworthy among the other specimens (although beginning to show their age) are two brides and bridegrooms. - Some information concerning these dressed fleas is contained in a letter, dated 12 October 1909, written by Mrs Zelia Nuttall (who between 1888 and 1928 published, in the U.S.A., various books and articles on ancient Mexico and Mexicans): "The inquiries I have made about the "pulgas vestidas" have yielded the following results: No one knows who first invented them, but they have "always been made" just like carved nuts and other curiosities. Those I sent you and all those supplied to the shop where I bought them are made by a middle-aged Hispano-Mexican woman - the mother of a large family - who lives in Querétaro. It seems that she often sits up at night so as to be undisturbed by her children and dress the fleas [Ctenocephalides felis and Pulex irritans] which they probably catch for her in the day-time ! Her latest productions are too funny for words. One flea represents a bull and another a toreador in front of it. I have got this group for you and will send it by the first opportunity. Of course I shall make further inquiries about the origin of these "curiosidades". I imagine they originated in some convent for the nuns excelled in making "miniaturas" of all kinds." - There are no further letters on this subject from Mrs Nuttall. - I understand that pulgas vestidas are still being made and sold in Mexico but now apparently without a real pulga, alas. Things surely are, or will be, no longer what they used to be . . .

F.G.A.M.S.